



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

**Salem Terminal TO MUTHUKUMAR & CO**

### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

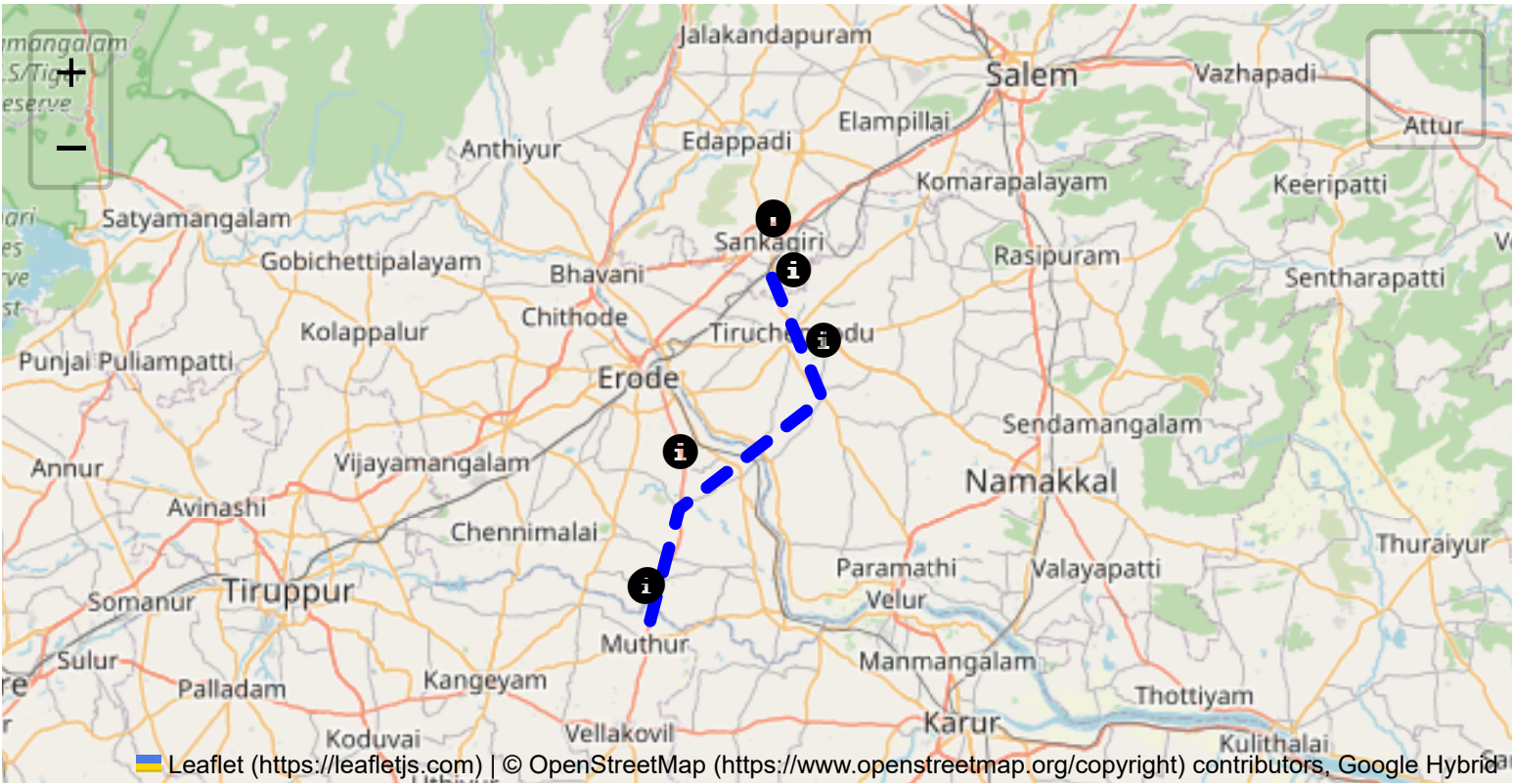
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

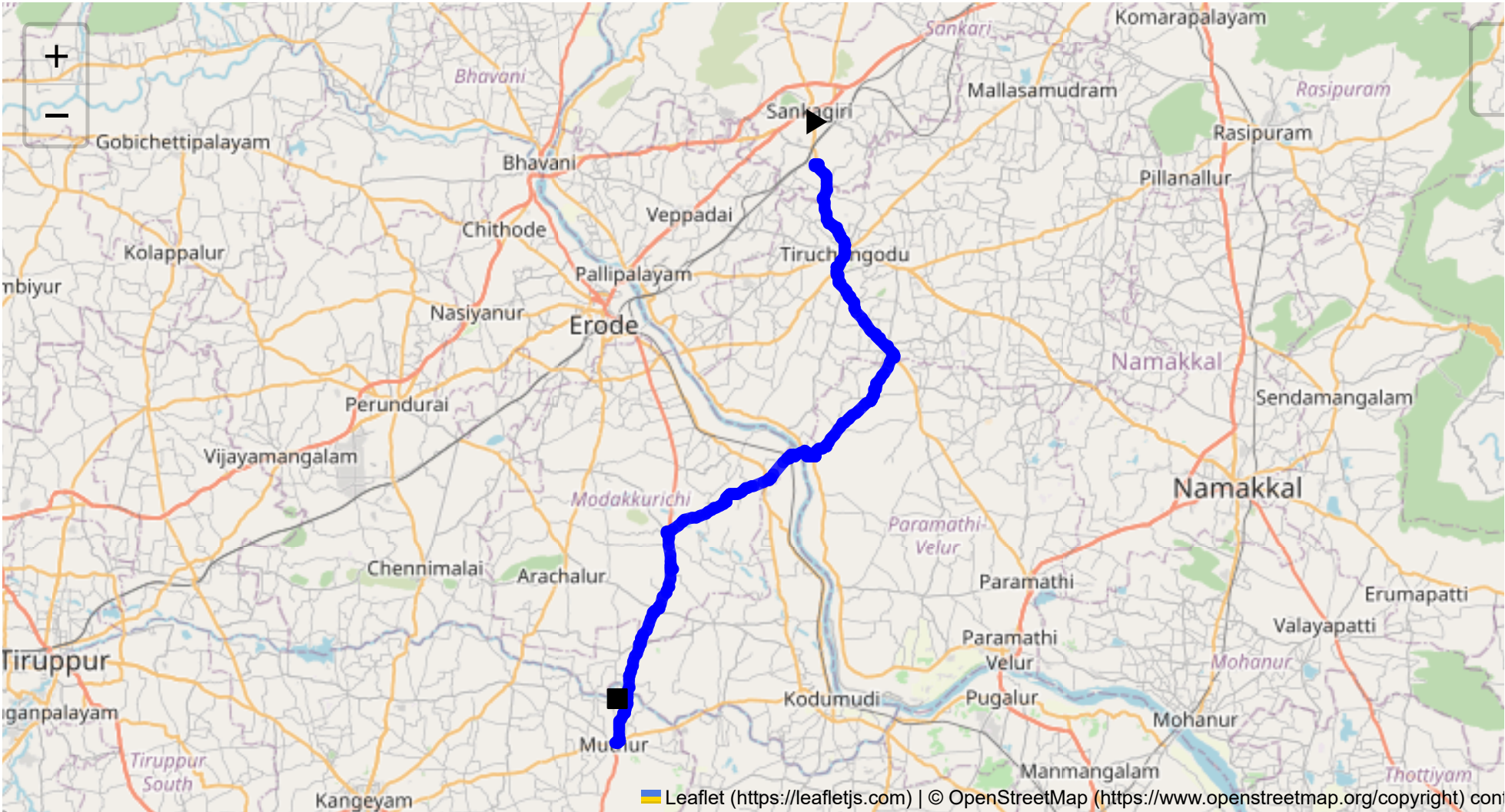
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 58.49 km**  
**Estimated Duration: 1.4 hours**  
**Adjusted Duration (Heavy Vehicle): 1.7 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.04468, 77.734535)**



**Risk Spots**

|   | RiskSpotType | Coordinates        | Speed Limit | Distance from Start |
|---|--------------|--------------------|-------------|---------------------|
| 2 | Turn         | 11.43822, 77.87348 | 15 KM/Hr    | 0.01 km             |
| 3 | Turn         | 11.43961, 77.87341 | 15 KM/Hr    | 0.14 km             |
| 4 | Blind Spot   | 11.44029, 77.87544 | 10 KM/Hr    | 0.31 km             |



|    | RiskSpotType | Coordinates        | Speed Limit | Distance from Start |
|----|--------------|--------------------|-------------|---------------------|
| 5  | Turn         | 11.39671, 77.88878 | 15 KM/Hr    | 5.84 km             |
| 0  | Roundabout   | 11.38395, 77.89480 | 15 KM/Hr    | 7.45 km             |
| 6  | Turn         | 11.37868, 77.89498 | 15 KM/Hr    | 7.97 km             |
| 7  | Turn         | 11.30859, 77.92918 | 15 KM/Hr    | 17.55 km            |
| 8  | Turn         | 11.24000, 77.87363 | 15 KM/Hr    | 27.57 km            |
| 9  | Turn         | 11.24053, 77.86902 | 15 KM/Hr    | 28.08 km            |
| 10 | Turn         | 11.24155, 77.86877 | 15 KM/Hr    | 28.20 km            |
| 11 | Turn         | 11.24248, 77.86829 | 15 KM/Hr    | 28.34 km            |
| 12 | Blind Spot   | 11.24384, 77.86689 | 10 KM/Hr    | 28.52 km            |
| 13 | Turn         | 11.24100, 77.86026 | 15 KM/Hr    | 29.30 km            |
| 14 | Turn         | 11.24194, 77.85910 | 15 KM/Hr    | 29.50 km            |
| 15 | Turn         | 11.23959, 77.85560 | 15 KM/Hr    | 29.98 km            |
| 16 | Blind Spot   | 11.18861, 77.77108 | 10 KM/Hr    | 41.16 km            |
| 17 | Turn         | 11.16417, 77.77367 | 15 KM/Hr    | 43.92 km            |
| 1  | Roundabout   | 11.04637, 77.73761 | 15 KM/Hr    | 58.00 km            |
| 18 | Turn         | 11.04515, 77.73621 | 15 KM/Hr    | 58.19 km            |

## Emergency Services

|    | type     | name                             | coordinates            | side_of_road  |
|----|----------|----------------------------------|------------------------|---------------|
| 0  | hospital | Tiruchengode, Goverment Hospital | 11.3903328, 77.8920627 | Driver's Side |
| 1  | hospital | SPM Medical Centre,Tiruchengode  | 11.3881331, 77.8931963 | Opposite Side |
| 2  | clinic   | Kongu Nursing Home               | 11.3783065, 77.8961134 | Driver's Side |
| 3  | hospital | T.C.A Hospital Tiruchengode      | 11.3791885, 77.8965774 | Driver's Side |
| 4  | hospital | Soorya Multispecialty Hospital   | 11.3786429, 77.8931912 | Opposite Side |
| 6  | hospital | Tiruchengode Government Hospital | 11.37645, 77.89426     | Driver's Side |
| 7  | hospital | Krishna Hospital, Namakkal       | 11.3754811, 77.8931817 | Opposite Side |
| 8  | hospital | Tirukumaran Hospitals            | 11.3782118, 77.8914933 | Opposite Side |
| 9  | police   | கரட்டுப்பாளையம் காவல் நிலையம்    | 11.357586, 77.8922539  | Driver's Side |
| 10 | hospital | Government Hospital, Elumathur   | 11.18533, 77.772496    | Driver's Side |
| 11 | hospital | Malar Hospital, Muthur           | 11.0438049, 77.7361156 | Driver's Side |

## Rest Stops

|    | type       | name       | coordinates            | side_of_road  |
|----|------------|------------|------------------------|---------------|
| 5  | restaurant | Lijo       | 11.3805115, 77.8946385 | Opposite Side |
| 12 | fuel       | Indian Oil | 11.0445808, 77.7345458 | Opposite Side |
| 13 | fuel       | Indian Oil | 11.0427213, 77.7366296 | Driver's Side |

## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route begins at CVQF+23W in Sangagiri and finishes at 2PVM+VR4 in Muthur, covering approximately 58.49 kilometers. It traverses through several key waypoints including Puthur, Thondikaradu near Girls High School, Sullipalayam on SH 86, Elumathur, and Kattupalayam.

### 2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate with hot summers, moderate monsoons, and mild winters. From June to September, heavy monsoonal rains can lead to waterlogging and reduced visibility on roads. Heat waves may occur from March to May, causing road surface degradation. Fog can be occasional during early mornings in winter (December to February).

### 3. Traffic Patterns

- Peak Hours:** Morning (8-10 AM) and evening (5-7 PM).
- Congestion-Prone Areas:** Sankagiri and Tiruchengode, particularly around school zones and marketplaces.
- Roads:** The SH 86 near Sullipalayam tends to have more traffic due to inter-district connectivity.

### 4. Assessment of Road Quality and Infrastructure

- Sangagiri to Puthur:** Moderate quality, with a few narrow sections.
- Thondikaradu:** Good, wider roads; speed limitations near schools.
- Sullipalayam (SH 86):** Good, with recent repairs visible but occasional potholes.
- Elumathur to Muthur:** Rural roads with varying quality, mostly single lane.

### 5. Suggestions for Alternative Routes for Emergencies

- SH 84A and SH 79:** Major highways with better facilities for rerouting in emergencies.
- Local Bypass Roads:** Can be used to detour major congestions, particularly around urban areas.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu adheres to national guidelines for transporting hazardous materials. Transport requires specific permits. Heavy vehicles must use designated routes, and nighttime driving might have restrictions due to safety concerns.

## 7. Overview of Historical Incidents

There have been sporadic incidents involving heavy vehicles on SH 86, mainly due to overspeeding and improper overtaking. Historical data indicate a few spills or leaks concerning hazardous materials due to inadequate container locking.

## 8. Environmental Considerations and Sensitive Areas

- **Agricultural Areas:** High sensitivity due to proximity to farmland.
- **Water Bodies:** Ensure spill prevention near water sources en route.
- **Schools and Hospitals:** Close proximity, especially near Thondikaradu, necessitates cautious driving and adherence to speed limits.

## 9. Analysis of Communication Coverage

Most of the route has adequate mobile network coverage. However, rural segments, particularly between Elumathur and Kattupalayam, may experience intermittent signal loss.

## 10. Estimated Emergency Response Times

- **Sangagiri to Tiruchengode:** 20-30 minutes (urban areas with proximal emergency services).
- **Tiruchengode to Sullipalayam:** 30-45 minutes.
- **Sullipalayam to Muthur:** 40-60 minutes, due to rural and less traversed areas.

## 12. Overall Summary of Risk Assessment

- **Low Risk:** Urban segments with good road infrastructure and emergency facilities.
- **Moderate Risk:** Rural areas with single-lane roads, potential for weather-related disruptions, and less frequent emergency services.
- **High Risk:** Peak hour travel through congested areas and weather conditions or incidents leading to road closures or diversions.

Safety protocols should emphasize cautious driving, speed management, and preparedness for weather disruptions. Emergency contact lists must be kept accessible, and planning should consider peak travel times and road quality variations. Training programs should focus on navigating rural terrains and handling hazardous materials securely.

## Route Photos of Risky Spots



**Risk Type:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Distance from Start:** 0.31 km

**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 5.84 km

**Coordinates:** 11.39671, 77.88878





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**Speed Limit:** 15 KM/Hr  
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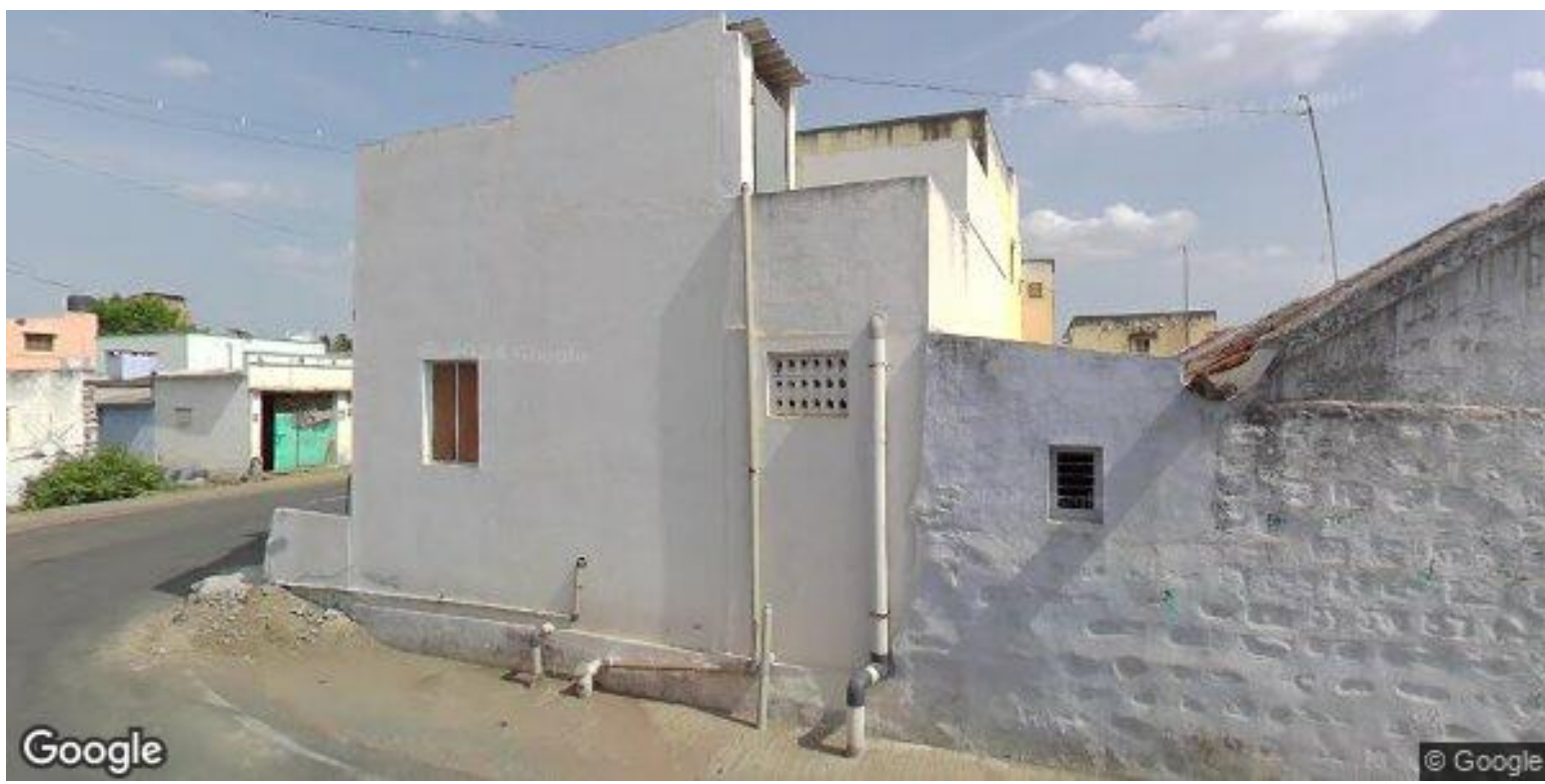
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**Speed Limit: 15 KM/Hr**

**Distance from Start: 28.08 km**

**Coordinates: 11.24053, 77.86902**





**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 28.20 km

**Coordinates:** 11.24155, 77.86877



**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 28.34 km

**Coordinates:** 11.24248, 77.86829





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**Speed Limit:** 15 KM/Hr

**Distance from Start:** 29.98 km

**Coordinates:** 11.23959, 77.85560



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**Coordinates:** 11.18861, 77.77108





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**Speed Limit: 15 KM/Hr**

**Distance from Start: 43.92 km**

**Coordinates: 11.16417, 77.77367**



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**Distance from Start: 58.00 km**

**Coordinates: 11.04637, 77.73761**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 58.19 km**

**Coordinates: 11.04515, 77.73621**

[Download Interactive Map](#)